What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Steroids

Last updated March 31, 2025 18 min read CC BY-SA 4.0

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Learning Objectives (6)

After completing this brick, you will be able to:

1. Explain the physiologic effects of cortisol and aldosterone.
2. Explain the mechanism of action of corticosteroids in relation to their clinical indications.
3. Classify select corticosteroids according to their effect, potency, and duration of action.
4. Predict corticosteroid adverse effects based on pharmacokinetics and pharmac
5. Explain short-term and long-term adverse effects of corticosteroids and discuss possible strategies for minimizing/preventing these adverse effects.
6. Justify when corticosteroids are contraindicated.

CASE CONNECTION

RL is a 45-year-old female who presents to the clinic with complaints of persistent fatigue, weight gain, and muscle weakness. She has a 5-year history of rheumatoid arthritis, managed with corticosteroids (prednisone 10 mg daily). Over the past year, she has also noticed that her blood pressure is harder to control (i.e., it is too high when she checks) and a rounding of her face. RL expresses concern about her appearance and struggles with her energy levels, which interfere with her ability to care for her family and work as a teacher.

How would you explain to RL why these effects have been occurring over the past year? Are there any strategies to reduce these effects? If she has to continue using corticosteroids, are there any other long-term effects she should watch out for?

GO TO CONCLUSION ↓

Before we take a look at the steroid medications, let's take a look at the body's primary endogenous steroids and their functions.

What do cortisol and aldosterone do? (LO#1)

The two most important endogenous steroids are cortisol and aldosterone. Cortisol is the primary glucocorticoid found in the body and is key in the regulation of blood pressure. Aldosterone is the main mineralocorticoid and it plays a critical role in maintaining blood volume.

The three organs involved in cortisol synthesis—hypothalamus, pituitary gland, and adrenal glands (HPA)—form the hypothalamic-pituitary-adrenal axis (HPA), which acts like a hormone-modulating center (Figure 1, left). First, the hypothalamus releases corticotropin-releasing hormone (CRH), which activates the adjacent anterior pituitary. The anterior pituitary then releases ACTH, which, in turn, triggers cortisol synthesis in the adrenal gland.

The HPA axis is responsible for increasing cortisol when it is needed, and it is responsible for stopping the production of more cortisol when there is too much. The hypothalamus releases CRH in a pulsatile way from hour to hour. This is how our body maintains an appropriate level of cortisol throughout the day. Although the level released may vary based on the time of day (Figure 1, right) and the level of stress someone is under, there are always some CRH, ACTH, and cortisol being produced. Note, the steps for aldosterone synthesis are the same as cortisol, however, it is released from the zona glomerulosa of the adrenal gland and, as mentioned above, has different physiologic functions as a mineralocorticoid.

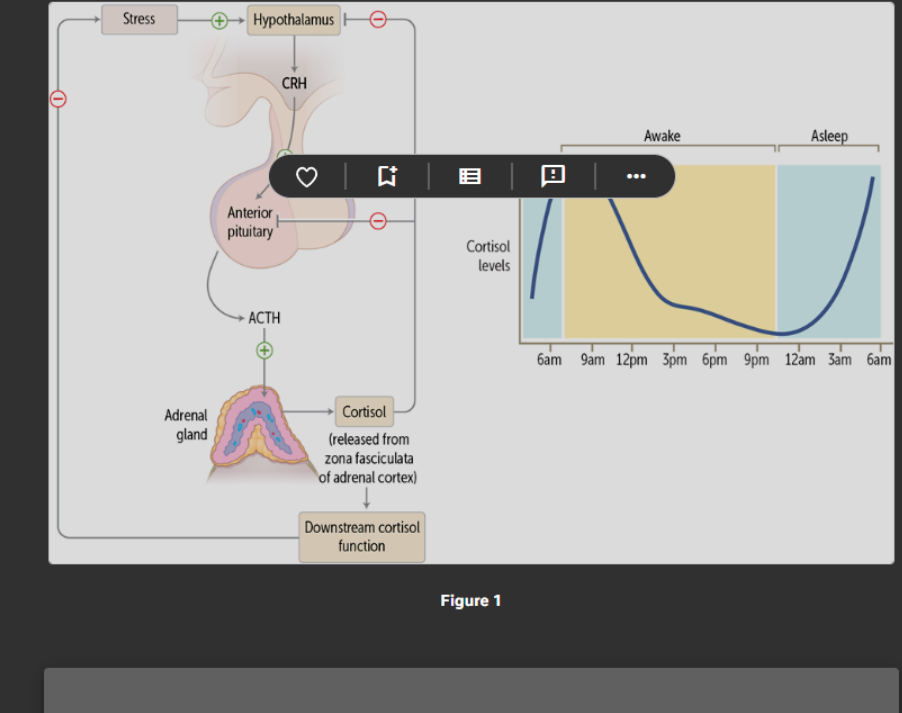


Figure 1

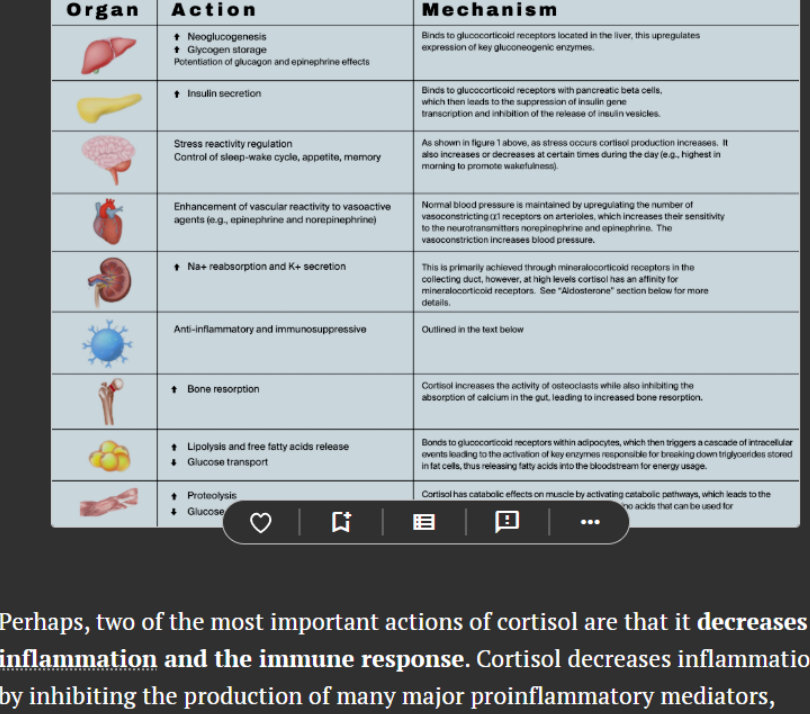
What hormone does the hypothalamus release to activate the anterior pituitary?

How do corticosteroids work? (LO#2)

Both cortisol and aldosterone have many different physiologic effects, and these effects are closely related to the clinical indications that the corticosteroid medications can treat. Common clinical indications are discussed below in conjunction with the effects and summarized in Table 1.

Cortisol (Glucocorticoid) Effects

While cortisol is important in maintaining blood pressure, it also has many other actions on many different areas of the body. These are included in the image below, but some of the most important effects (i.e., Anti-inflammatory and immunosuppression) are described in more detail below the image.



Actions of Cortisol

Organ	Action	Mechanism
	Neoglucogenesis Glycogen storage Potentiation of glucagon and epinephrine effects	[illegible]
	Insulin secretion	[illegible]
	Stress reactivity regulation Control of sleep-wake cycle, appetite, memory	[illegible]
	Enhancement of vascular reactivity to vasoactive agents (e.g., epinephrine and norepinephrine)	[illegible]
	Na+ reabsorption and K+ secretion	[illegible]
	Anti-inflammatory and immunosuppression	Outlined in the text below
	Bone resorption	[illegible]
	Lipolysis and free fatty acids release Glucose transport	[illegible]
	Proteolysis	[illegible]

Perhaps, two of the most important actions of cortisol are that it **decreases inflammation and the immune response.** Cortisol decreases inflammation by inhibiting the production of many major proinflammatory mediators, including leukotrienes, prostaglandins, interleukin 2 (IL-2), interferon γ (IFN-γ), IFN-α, and tumor necrosis factor α (TNF-α). Cortisol also increases the production of many anti-inflammatory mediators, including IL-10, IL-4, and IL-13. By reducing the number of inflammatory mediators in the upper dermis (i.e., with topical corticosteroids) it causes vasoconstriction and reduces inflammation, along with the shear reduction in cytokines reducing the allergic reaction. In a similar way, cortisol decreases the immune response (i.e., or can cause immunosuppression at high enough levels, like when giving corticosteroid medications) by suppressing both B- and T-cell function by decreasing transcription of many cytokines (like those mentioned above). Cortisol also induces T cell apoptosis which further reduces the immune response. Lastly, cortisol further decreases the immune response and impairs wound healing by decreasing the activity of fibroblasts and mast cells.

CLINICAL CORRELATION

Critically ill patients with severe unexplained hypotension, despite adequate fluid and vasopressor therapy, are often treated with hydrocortisone (i.e., pharmacologic cortisol). This can be a life saving measure as these patients may have a cortisol deficiency that is causing their hypotension.

Aldosterone (Mineralocorticoid) Effects

Aldosterone regulates salt and water in the body and thereby helps regulate blood volume and blood pressure. It achieves this by its actions in the kidneys. It is also involved in mechanisms outside the kidney.

The aldosterone receptor, or mineralocorticoid receptor (MR), acts as a ligand-dependent transcription factor (TF), meaning that when aldosterone binds the receptor, it acts as a TF. As a TF, the MR alters cell signaling in a variety of tissues in order produce proteins that regulate water and ion transport. Interestingly, the MR has similar affinity for cortisol and aldosterone but is "protected" from cortisol by an enzyme called corticosteroid 11β-dehydrogenase isozyme 2. This enzyme metabolizes cortisol to an inactive form when it tries to bind the MR. This type of regulation is crucial, because circulating cortisol concentrations are 100 times greater than circulating aldosterone concentrations; otherwise, cortisol would dominate MR signaling.

The primary function of aldosterone is to maintain blood volume and blood pressure through regulation of sodium reabsorption in the distal nephron of the kidney, specifically in the collecting duct (Figure 2). Recall that sodium reabsorption drives concurrent water reabsorption, so the net effect is to increase plasma volume. Sodium reabsorption is a key component of the blood pressure regulation.

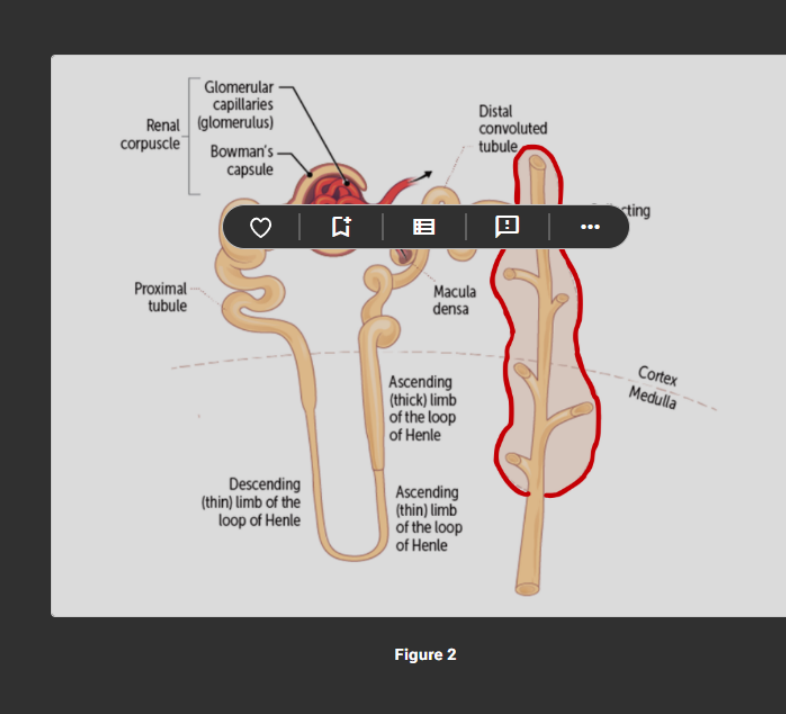

Glomerular capillaries (glomerulus)
Renal corpuscle
Bowman's capsule
Distal convoluted tubule
Proximal tubule
Macula densa
Ascending (thick) limb of the loop of Henle
Descending (thin) limb of the loop of Henle
Ascending (thin) limb of the loop of Henle
Cortex
Medulla

Figure 2

The net effect of actions in the collecting duct is sodium reabsorption and potassium excretion. Specifically:

- Aldosterone increases sodium reabsorption by increasing both the sodium channels on the lumen side of the cell and the Na+/K+-ATPase pump activity on the blood (basolateral) side of the cell.
- Stimulating the Na+/K+-ATPase pump also increases entry of potassium into the cell from the blood. Increasing the number of potassium channels then allows potassium to exit the cell and be secreted into the urine. This is how aldosterone increases urinary potassium excretion.

While the kidney is the main target of aldosterone, the hormone also stimulates sodium reabsorption through its actions on receptors in the colon, salivary, and sweat glands, assisting the kidney with expansion of plasma volume.

Table 1 - Common Clinical Indications

Glucocorticoids	Mineralocorticoids
Autoimmune and inflammatory disorders (e.g., rheumatoid arthritis)	Adrenal insufficiency (i.e., Addison's disease)
Adrenal insufficiency	Cerebral salt wasting
Asthma Exacerbation	Adrenogenital disorder
Enhance fetal pulmonary maturity (given IM antenatal)	Postural orthostatic tachycardia syndrome
Post-transplant rejection	
Cancer (e.g., non-Hodgkin lymphoma)	
Skin Conditions (e.g., Eczema and Psoriasis)	

CLINICAL CORRELATION

Hypertension can be caused by primary hyperaldosteronism. In this disorder, the adrenal glands continuously secrete aldosterone without regulation. The result is sustained activation of the aldosterone-dependent pathways, causing sodium and water retention, hypertension, and profound hypokalemia caused by urinary potassium wasting.

Patients with this disorder will have persistent hypertension, despite adequate treatment with 3-4 blood pressure medications. Adding an aldosterone antagonist (i.e., spironolactone) can reduce aldosterone levels and effectively lower blood pressure.

How does cortisol decrease inflammation?

CLINICAL CORRELATION

Intramuscular antenatal steroids should be given to enhance fetal pulmonary maturity. In fact, the use of antenatal corticosteroids is the single most important obstetrical intervention available that positively impacts neonatal morbidity and survival.

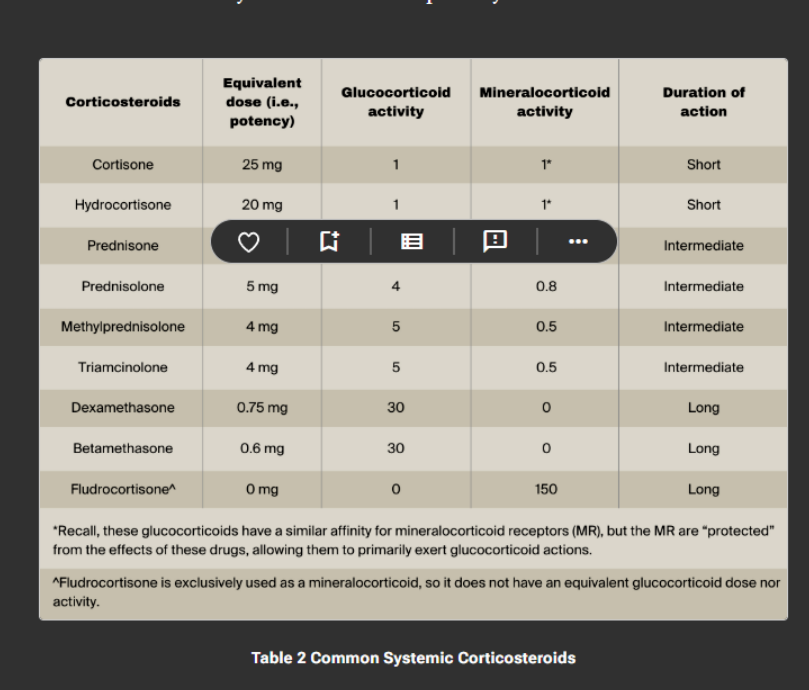
Antenatal steroids, given IM (Betamethasone 12 mg (2 doses 24 hours apart) for preterm labor occurring between 23rd and 36th weeks, accelerates development of type 1 and type 2 pneumocytes, leading to structural and biochemical changes that improve both lung mechanics and gas exchange (eg, surfactant production).

How are cortic)

Like many other medication classes, each corticosteroid medication has differences in their range of glucocorticoid vs mineralocorticoid activity, effects, potency, and duration of action. Table 2 compares the pharmacology of common corticosteroid medications.

Much of the action of each individual medication is related to how much, or how little, glucocorticoid activity it exhibits. Those corticosteroids with high glucocorticoid activity, like dexamethasone, have a high potency (e.g. higher affinity than cortisol to suppress proinflammatory mediators). However, potency can also be greatly influenced by the route of administration. Take topical corticosteroids for instance. You may have varying concentrations, or strengths, available of the same corticosteroid. The area that is being treated, i.e., where the steroid is applied, can greatly increase the realized potency of the medication. An area with thinner skin will require lower potency steroids, and if an occlusive bandage is applied it will increase the absorption.

Duration of each medication is another important point to keep in mind. It can be affected by the structure of the medication and/ or the formulation. Those steroids that are more long-acting, like dexamethasone, may be advantageous for a pediatric asthma exacerbation, as it would only have to be dosed once and exert anti-inflammatory effects for multiple days.



Corticosteroids	Equivalent dose (i.e., potency)	Glucocorticoid activity	Mineralocorticoid activity	Duration of action
Cortisone	25 mg	1	1*	Short
Hydrocortisone	20 mg	1	1*	Short
Prednisone				Intermediate
Prednisolone	5 mg	4	0.8	Intermediate
Methylprednisolone	4 mg	5	0.5	Intermediate
Triamcinolone	4 mg	5	0.5	Intermediate
Dexamethasone	0.75 mg	30	0	Long
Betamethasone	0.6 mg	30	0	Long
Fludrocortisone*	0 mg	0	150	Long

*Recall, these glucocorticoids have a similar affinity for mineralocorticoid receptors (MR), but the MR are "protected" from the effects of these drugs, allowing them to primarily exert glucocorticoid actions.

*Fludrocortisone is exclusively used as a mineralocorticoid, so it does not have an equivalent glucocorticoid dose nor activity.

Table 2 Common Systemic Corticosteroids

CLINICAL CORRELATION

For topical corticosteroids in pediatric patients, lower concentrations should be used as the potential for them to have significant systemic absorption is higher. This is because pediatrics have larger surface area to body mass ratio, and, more importantly, they have thinner skin than adults. Neonates are at the highest risk for systemic absorption and the risk steadily decreases until a patient reaches 12 years old. At this age the risk for systemic absorption is equivalent to adults.

Why might dexamethasone be preferred for pediatric asthma exacerbations?

What are the adverse effects of steroids? (LO#4)

Corticosteroids are a medication class with many important side effects that should be taken into consideration before they are prescribed. Often, the side effects are related to the level of glucocorticoid effects they have. This section of the brick will discuss the most common general adverse effects related to the pharmacokinetics/ pharmacodynamics, in the next section we will distinguish the short- and long-term effects, as well as strategies to mitigate those effects.

Adrenal suppression: Endogenously, the production of corticosteroids is under the regulation of the hypothalamic-pituitary-adrenal (HPA) axis. The rapid induction of corticosteroids occurs in response to inflammation and other stressors, but they also follow secretion patterns associated with circadian rhythms. Adrenal suppression occurs when the exposure of the HPA axis to exogenous corticosteroids leads to inadequate cortisol production. Importantly, the duration and dosage of corticosteroid therapy are not reliable predictors of when patients will develop suppression of their HPA axis. If enough systemic abso ical corticosteroids can cause adrenal supp often occur after **sudden discontinuation of corticosteroid treatment,** and therefore, gradual tapering is often part of corticosteroid treatment protocols. See clinical correlation below for more on this topic.

Diabetes and hyperglycemia: Corticosteroids are the most common cause of drug-induced diabetes mellitus. As corticosteroids mimic cortisol this initiates increased gluconeogenesis from the liver, it also decreases insulin synthesis from the pancreas. This process can cause new onset or worsen current diabetes in a patient. A dose-dependent hyperglycemic effect occurs in patients within hours of exposure.

Gastrointestinal adverse effects: Multiple gastrointestinal effects correlate with corticosteroid therapy, including gastritis, peptic ulcer disease, abdominal distention, and dyspepsia. The risk increases significantly, nearly four-fold, when used in combination with nonsteroidal anti-inflammatory drugs (NSAIDs). The exact mechanism is not fully understood, but it likely has to do with the delayed wound healing, as well as increased stomach acid secretion.

Topical corticosteroids have their own set of side effects, while, generally, the systemic side effects (e.g., adrenal suppression) do not occur they can still occur if enough is absorbed systemically. The topical side effects associated with corticosteroids include:

- Skin irritation
- Skin thinning
- Stretch marks
- Acne
- Rosacea
- Skin color changes
- Excessive hair gr
- Worsening or spreading of an existing skin infection
- Inflamed hair follicles (folliculitis)
- Easy bruising

CLINICAL CORRELATION

A patient becomes immunosuppressed from steroids when using ≥ 2 mg/kg/day or **≥ 20 mg/day** of prednisone or prednisone equivalent (recall equivalent doses from Table 2 above) for **> 2 weeks**.

For these patients, taper steroids when discontinuing to avoid adrenal suppression. There are many ways to taper steroids, but one of the most common methods is to reduce the dose about 10-20% every few days. Taper lengths will vary depending on the condition and total daily dose.

What increases the risk of gastrointestinal effects when using corticosteroids?

What are the short- and long-term effects of corticosteroids? (LO#5)

Often, the adverse effects of corticosteroids are grouped by short- and long-term effects. For clinical indications, like an asthma exacerbation, only a short course (≤ 5 days) is warranted and some of the adverse effects are avoided. With long-term use, like for immunosuppression in a transplant rejection, a myriad of adverse effects come in to play.

Even with short-term corticosteroid use there are still adverse effects to consider. Some of the most pertinent include:

- *Hyperglycemia* - Recall from above, a dose-dependent hyperglycemic effect occurs in patients within hours of exposure.
- *Psychiatric disturbances* - While this psychiatric disturbances are more common with extended courses, short-term corticosteroid use can cause, or exacerbate an underlying, a range of psychiatric disorders, including psychosis, agitation, insomnia, irritability, hypomania, anxiety, and mood lability. Also, short courses of corticosteroids can produce euphoria in many individuals.
- *Cardiovascular effects* - In patients with history of cardiac disorders, even short-term corticosteroid use can induce hypertension, atrial fibrillation or flutter, or heart failure exacerbations.
- *Oral Thrush (Inhaled corticosteroids)* - Inhaled corticosteroids are a mainstay treatment for asthma and COPD. They can increase, especially in children, the risk of thrush, an oral fungal infection due to *Candida albicans*. As we h en the immune response in the mouth and throat opportunistic fungi can cause thrush.

The long-term effects are summarized in Figure 4 below.

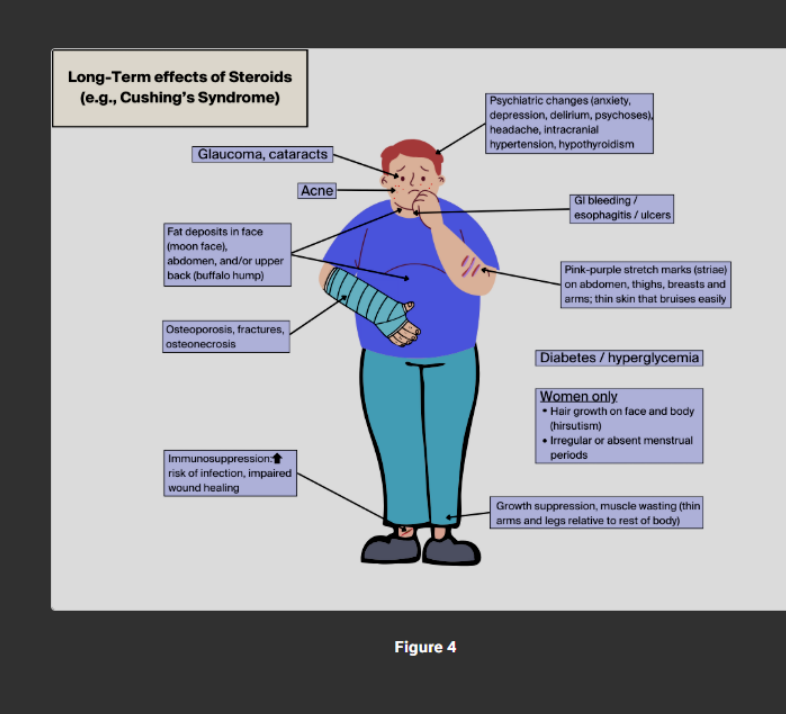

Long-Term effects of Steroids (e.g., Cushing's Syndrome)
Glaucoma, cataracts
Acne
Fat deposits in face (moon face), abdomen, and/or upper back (buffalo hump)
Osteoporosis, fractures, osteonecrosis
Immunosuppression ↑ risk of infection, impaired wound healing
Psychiatric changes (anxiety, depression, delirium, psychosis), headache, intracranial hypertension
GI bleeding / esophagitis / ulcers
Pink-purple stretch marks (striae) on abdomen, thighs, breasts and arms, thin skin that bruises easily
Diabetes / hyperglycemia
Women only
• Hair growth on face and body (hirsutism)
• Irregular or absent menstrual periods
Growth suppression, muscle wasting (thin arms and legs relative to rest of body)

Figure 4

Some diseases require steroid therapy for effective treatment. Some strategies to reduce systemic steroid effects include:

- Use every other day dosing, this can reduce adrenal suppression
- Intra-articular injections, rather than systemic injections
- For asthma treatment, use inhaled corticosteroids that mainly stay in the lungs and reduce rush, *the patient should rinse their mouth immediately each time after using their inhaler.*
- Use the lowest effective dose for the shortest possible time

What is a strategy to reduce systemic effects of steroids in asthma treatment?

What are the contraindications for corticosteroids? (LO#6)

With the effects of corticosteroids on the immune system, many of the contraindications are related to that side effect. Contraindications to corticosteroids include:

- Concurrent administration of live or live-attenuated vaccines (when using immunosuppressive dosages)
- Systemic fungal infection
- Osteoporosis
- Uncontrolled hyperglycemia
- Diabetes mellitus
- Glaucoma
- Joint infection
- Uncontrolled hypertension
- Herpes simplex keratitis
- Varicella infection
- Peptic ulcer disease (or active GI bleed)
- Congestive heart failure
- Viral or bacterial infections not controlled by anti-infectives

CLINICAL CORRELATION

What would happen if you give a live, or live-attenuated vaccine, to patient on a corticosteroid? What about giving a corticosteroid to a patient with diabetes, or peptic ulcer disease?

To answer these questions we have to consider the side effects of corticosteroids (which are many!). Let's consider the three situations we mentioned here. First, remember that corticosteroids, long-term and high-doses, will cause immunosuppression. Giving a live vaccine to this patient will be ineffective because they do not have the ability to mount the appropriate immune response. Corticosteroids increase blood sugar and carry a risk for GI bleed. So, a patient with diabetes will need to monitor their blood sugar more closely while on corticosteroids, and, potentially, avoiding steroids in patients with a history of ulcers (i.e., GI bleeds) would be good practice.

Why are live vaccines contraindicated with corticosteroids?

ACTIVE LEARNING QUESTION

Which hormone is released by the hypothalamus to initiate cortisol synthesis?

A. ● CRH
B. ○ ACTH
C. ○ Cortisol
D. ○ Aldosterone

Explanation

The correct answer is CRH (A). The hypothalamus releases corticotropin-releasing hormone (CRH), which activates the anterior pituitary. ACTH (B) is released by the anterior pituitary, not the hypothalamus. Cortisol (C) is synthesized in the adrenal gland. Aldosterone (D) is a mineralocorticoid, not involved in this specific process.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking about RL and her long-term corticosteroid use. How would you explain to her why these effects are happening? Can she do anything to mitigate these effects, and is there anything else she might need to watch out for?

Sometimes, long-term corticosteroid use is unavoidable to treat painful conditions, like rheumatoid arthritis in the case of RL. Following expression of empathy and concern for the difficult situation she finds herself in you might briefly and plainly explain that corticosteroids, like prednisone suppress the immune system. This is necessary with a condition like rheumatoid arthritis. Taking exogenous steroids decreases the activity of the cells involved with immune response, but by adding these medications it signals the body to stop producing its own natural steroids (i.e., cortisol). This medication also affects many other organs of the body, like the heart/ blood vessels, liver, pancreas, bones, muscles, an ensitivity to certain chemicals within the body that can increase blood pressure, making it more difficult to control. Further, it can increase fat deposits in the face, causing a 'moon face' appearance. One thing to consider for RL would be to reduce the dose and via taper regimen. If her arthritis symptoms remain controlled, or if there are other medication options, then this could help reduce some of the effects. If she has to continue steroids, then she would need to monitor for diabetes and osteoporosis along with some other long-term side effects.

Summary

- Cortisol and aldosterone are the body's primary endogenous steroids, with cortisol playing a key role in blood pressure regulation and aldosterone maintaining blood volume.
- The hypothalamic-pituitary-adrenal (HPA) axis is responsible for regulating cortisol synthesis, with the hypothalamus releasing corticotropin-releasing hormone (CRH) to activate the anterior pituitary, which then releases ACTH to trigger cortisol synthesis in the adrenal gland.
- Cortisol and aldosterone have various physiological effects that are closely related to the clinical indications that corticosteroid medications can treat, such as anti-inflammatory and immunosuppressive effects.
- Corticosteroids are classified based on their range of glucocorticoid vs mineralocorticoid activity, effects, potency, and duration of action.
- Corticosteroids have numerous adverse effects, including adrenal suppression, diabetes and hyperglycemia, gastrointestinal adverse effects, and various topical side effects.
- Contraindication t administration of live or live-attenuated vaccines, systemic fungal infection, osteoporosis, uncontrolled hyperglycemia, diabetes mellitus, glaucoma, joint infection, uncontrolled hypertension, herpes simplex keratitis, varicella infection, peptic ulcer disease, and congestive heart failure.

Review Questions

1. Which drug class is most commonly associated with drug-induced diabetes mellitus?

A. ● Corticosteroids
B. ○ NSAIDs
C. ○ Antihistamines
D. ○ Antibiotics
E. ○ Beta-blockers

Hide Explanation

The correct answer is corticosteroids (A). Corticosteroids mimic cortisol, leading to increased gluconeogenesis and decreased insulin synthesis, causing hyperglycemia. NSAIDs (B) are associated with gastrointestinal adverse effects but not directly with diabetes. Antihistamines (C) and antibiotics (D) do not have a significant impact on glucose metabolism. Beta-blockers (E) can affect glucose levels but are not the

2. Which of the following best describes the mechanism by which corticosteroids decrease inflammation?

A. ○ Increase proinflammatory mediators
B. ● Inhibit proinflammatory mediators and stimulate anti-inflammatory mediators
C. ○ Decrease anti-inflammatory mediators
D. ○ Inhibit both proinflammatory and anti-inflammatory mediators
E. ○ Stimulate both proinflammatory and anti-inflammatory mediators

Hide Explanation

The correct answer is inhibition of proinflammatory mediators and stimulation of anti-inflammatory mediators (B). Corticosteroids decrease inflammation by inhibiting the production of many major proinflammatory mediators, including leukotrienes, prostaglandins, interleukin 2 (IL-2), interferon γ (IFN-γ), IFN-α, and tumor necrosis factor α (TNF-α). Corticosteroids also increase the production of many anti-inflammatory mediators, including IL-10, IL-4, and IL-13. The other options are incorrect. Corticosteroids do not increase proinflammatory mediators (A), they do not decrease anti-inflammatory mediators (C), they do not inhibit both proinflammatory and anti-inflammatory mediators (D), and they do not stimulate both proinflammatory and anti-inflammatory mediators (E).

3. Which of the following corticosteroids has the highest glucocorticoid activity and longest duration of action?

A. ○ Cortisone
B. ○ Hydrocortisone
C. ○ Prednisone
D. ● Dexamethasone
E. ○ Fludrocortisone

Hide Explanation

The correct answer is Dexamethasone (D). Dexamethasone has a high glucocorticoid activity of 30 and a long duration of action. Cortisone (A) and Hydrocortisone (B) have a glucocorticoid activity of 1 and a short duration of action. Prednisone (C) has a glucocorticoid activity of 4 and an intermediate duration of action. Fludrocortisone (E) is used exclusively as a mineralocorticoid and does not have glucocorticoid activity.

4. Which of the following is a contraindication for corticosteroids?

A. ○ Diabetes mellitus
B. ● Systemic
C. ○ Glaucoma
D. ○ Uncontrolled hypertension
E. ○ Osteoporosis

Hide Explanation

The correct answer is systemic fungal infection (B). Corticosteroids suppress the immune system, making it harder for the body to fight off infections, including fungal infections. While corticosteroids can exacerbate conditions like diabetes mellitus (A), glaucoma (C), and hypertension (D), these are not absolute contraindications. Corticosteroids are not contraindicated in osteoporosis (E), but they can worsen the condition and should be used with caution.

5. Which of the following strategies can be used to reduce systemic effects of corticosteroids in asthma treatment?

A. ○ Every other day dosing
B. ● Using inhaled corticosteroids
C. ○ Lowest effective dose for shortest time
D. ○ Intra-articular injections
E. ○ Systemic

Hide Explanation

The correct answer is using inhaled corticosteroids (B). Inhaled corticosteroids mainly stay in the lungs and reduce systemic absorption, thereby minimizing systemic effects. Using every other day dosing (A) can reduce adrenal suppression but is not specific to asthma treatment. While using the lowest effective dose for the shortest possible time (C) is a general strategy for reducing adverse effects of corticosteroids, it does not specifically address reducing systemic effects in asthma treatment. Intra-articular injections (D) are not applicable for asthma treatment. Systemic injections (E) can actually increase systemic effects.

6. Which hormone is released by the hypothalamus to initiate cortisol synthesis?

A. ● CRH
B. ○ ACTH
C. ○ Cortisol
D. ○ Aldosterone

Hide Explanation

The correct answer i cotropin-releasing hormone (CRH), which activates the anterior pituitary. ACTH (B) is released by the anterior pituitary, not the hypothalamus. Cortisol (C) is synthesized in the adrenal gland. Aldosterone (D) is a mineralocorticoid, not involved in this specific process.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the physiologic effects of cortisol and aldosterone.~~
02 ☑ ~~Explain the mechanism of action of corticosteroids in relation to their clinical indications.~~
03 ☑ ~~Classify select corticosteroids according to their effect, potency, and duration of action.~~
04 ☑ ~~Predict corticosteroid adverse effects based on pharmacokinetics and pharmacodynamics.~~
05 ☑ ~~Explain short-term and long-term adverse effects of corticosteroids and discuss possible strategies for minimizing/preventing these adverse effects.~~
06 ☑ ~~Justify~~

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Objective Confidence: 6/6